

SUNNY SHAH

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Education

University of Michigan

Expected May 2027

Bachelor of Science in Computer Science and Pure Mathematics

Ann Arbor, MI

- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Computer Networks, Distributed Systems, Computer Organization, Web Systems, Machine Learning, Probability Theory, Discrete Mathematics, Object-Oriented Programming

Experience

Software Engineer Intern

May 2026 – Present

Capital One

Richmond, VA

- Automate credit-card test-data provisioning for QA engineers across 5+ teams by building a Claude Skill that orchestrates Python tooling, Test Data Management APIs, downstream service updates, and Playwright validation end to end
- Achieve 90% HitRate@5 on a labeled evaluation set and reduce Claude prompt context by approximately 10,000 tokens by developing a BM25-based lexical and structural retrieval engine over 600+ test recipes and 1,200+ existing tests
- Prevent stale or inconsistent test state before UI execution by implementing cross-system validation and fail-fast checks across TDM provisioning, downstream service updates, and Playwright tests

Data & Artificial Intelligence Engineering Intern

May 2025 – August 2025

Protiviti

Chicago, IL

- Built a Streamlit FP&A platform that ingested eight enterprise ledger sources, validated schemas, coerced data types, and imputed missing values, producing model-ready datasets for 100,000+ financial records
- Achieved 90% category accuracy and sub-10% 12-month forecast MAPE by engineering features and training Random Forest, Prophet, and Isolation Forest models, surfacing top-25 anomalies and live predictions in executive dashboards
- Tripled ingestion throughput using PyArrow parsing and session-state caching while generating fully numeric training matrices in under two seconds on cloud-hosted VDI hardware

Software Engineer Intern, Research

November 2023 – August 2024

Wayne State University – College of Engineering

Detroit, MI

- Reduced analysis time across 20+ simulation datasets by vectorizing NumPy workflows and building Power BI and Python dashboards to compare NVIDIA Omniverse drone-design experiments; secured \$1,000+ in project funding through an authored research proposal

Projects

Sharded Paxos Key-Value Store | *Go, Paxos, RPC, Distributed Systems*

April 2026

- Built a sharded, fault-tolerant key-value service in Go exposing Get, Put, and Append RPCs, distributing data across replica groups while preserving linearizability under concurrent requests and server failures
- Implemented Paxos consensus and a replicated state-machine log for deterministic operation ordering; designed a Paxos-backed shard master that rebalanced ownership and coordinated shard migration as cluster configurations changed

Multi-Threaded Network File Server | *C++20, TCP Sockets, Boost, RAII*

December 2025

- Built a multithreaded C++ network file server with inode- and directory-entry-based storage, enabling concurrent remote clients to create, read, write, and delete files over TCP sockets
- Maximized safe request concurrency with per-inode reader-writer locks, Boost shared, upgradeable, and unique locks, and hand-over-hand locking while preserving atomic filesystem updates without races or deadlocks

User-Level Thread Library | *C++20, ucontext, Mutexes, Condition Variables*

October 2025

- Implemented a user-level threading runtime supporting thread creation, joining, preemption, mutexes, and condition variables across multi-CPU systems using FIFO scheduling, ucontext switching, timer and inter-processor interrupts, and atomic interrupt control

Adaptive HTTP Proxy & CDN Load Balancer | *C++17, HTTP, TCP, Poll*

October 2025

- Built a non-blocking C++ HTTP proxy and CDN load balancer that rewrote MPEG-DASH requests using per-client throughput estimates and routed traffic with round-robin and geographic policies, improving playback stability under fluctuating bandwidth

IMC Prosperity Trading Competition | *Python, pandas, NumPy, scikit-learn*

April 2025

- Ranked in the top 5% of 13,000+ competitors by developing a multi-strategy trading bot combining market making, mean reversion, momentum, volatility modeling, and synthetic-basket arbitrage under liquidity and position constraints

Technical Skills

Languages: C++20, Go, Python, SQL, JavaScript (TypeScript), Bash, HTML/CSS

Frameworks: React, Flask, Streamlit, Playwright, PyTorch

Libraries: STL, Boost, pandas, NumPy, Matplotlib, SciPy, scikit-learn

Systems & Tools: Linux, Amazon Web Services (AWS), Google Cloud Platform (GCP), Docker, Git, TCP/IP, POSIX Sockets, Mininet, PostgreSQL, VS Code, GDB, CMake, Codex CLI, Claude Code